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2nd Sem / Electrical

Subject : Non-Conventional Sources of Energy

Time : 3 Hrs.

M.M. : 60

SECTION-A

Note: Multiple choice questions. All questions are compulsory (6x1=6)

Q.1 MHD stands for _____

- a) Magneto hydrogen dynamic
- b) Magneto hydro dynamic
- c) Magnetic hydrogen dynamic
- d) None of these

Q.2 The hydro power is a clean, natural source of _____ energy.

- a) Sun
- b) Water
- c) Wind
- d) Gas

Q.3 Solar cells provides _____ output

- a) DC
- b) AC
- c) Both DC & AC
- d) None of these

Q.4 Which of these is a non-renewable source of energy

- a) Biomass b) Solar
- c) Coal d) Wind

Q.5 Geothermal energy is obtained from

- a) Light b) Solar
- c) Wind d) Earth

Q.6 Fuel cell converts _____ into _____

- a) Chemical, Electrical
- b) Electrical, Chemical
- c) Mechanical, Chemical
- d) None of these

SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (6x1=6)

Q.7 Write the types of wind turbine.

Q.8 Define solar panel.

Q.9 Define non-conventional sources of energy.

Q.10 List different sources of biomass.

Q.11 Write the name of electrodes of fuel cell.

Q.12 Write full form of OTEC.

SECTION-C

Note: Short answer type questions. Attempt any eight questions out of ten questions. (8x4=32)

- Q.13 What are the advantages and disadvantages of renewable energy sources?
- Q.14 Describe the principle of photovoltaic energy conversion.
- Q.15 What is hydro energy? How it can be converted into useful forms?
- Q.16 Explain different modes of energy storage system.
- Q.17 Write some advantages of biogas plants.
- Q.18 State and explain the principle of working of fuel cell.
- Q.19 Explain close cycle system OTEC.
- Q.20 What are the parts of tidal energy plant?
- Q.21 State the factors for selecting wind mill site.
- Q.22 Describe geothermal energy. Why it is called renewable energy?

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x8=16)

Q.23 Compare conventional energy sources and non-conventional energy sources. Also compare standalone PV system with grid connected PV system.

Q.24 Explain the various types of tidal power plants.

Q.25 What is gasifier? Explain various types of gasifier.